

Threat Hunting is a **proactive cybersecurity process** where security analysts actively search inside an organization's network to find hidden threats that security tools (SIEM, antivirus, EDR) may have missed.

It is **not waiting** for alerts —

It is **actively hunting attackers** who may already be inside the network.

✅ Simple Definition

Threat Hunting is the process of searching for abnormal or malicious activity inside systems before it causes damage.



Why Threat Hunting is needed?

Because:

- Attackers bypass antivirus
- Malware hides or runs fileless
- SIEM alerts can miss new attack patterns
- Insider threats cannot be detected by tools
- Zero-day attacks have no signatures

Threat hunters look for **patterns, anomalies, TTPs (MITRE ATT&CK)** to detect early signs of compromise.



What do Threat Hunters look for?

They search for:

- Unusual login locations
- Abnormal network traffic
- Suspicious PowerShell or CMD commands
- Connections to known malicious IPs
- Persistence mechanisms
- Lateral movement
- Data exfiltration patterns



Threat Hunting Phases

1. Hypothesis creation

Example:

“Is there any suspicious PowerShell usage in the last 7 days?”

2. Data collection + logs

Collect logs from:

- SIEM
- EDR
- Firewall

- DNS logs
- Endpoint logs

3. Investigation

Analyze:

- MITRE ATT&CK behavior
- IOCs
- TTPs
- Suspicious patterns

4. Detection

If malicious activity is found:

- Mark IOC
- Write detection rule
- Block attacker

5. Response

Take action:

- Disable user
- Block IP/domain
- Quarantine endpoint
- Patch vulnerability

Goal of Threat Hunting

- Catch hidden threats early
- Reduce dwell time
- Improve SOC security
- Create new SIEM detection rules
- Strengthen organization's defense

Threat Hunting Tools

- SIEM (Splunk, QRadar, Sentinel)
- EDR (CrowdStrike, SentinelOne)
- ELK / Wazuh
- MITRE ATT&CK
- OTX, VirusTotal, Shodan
- Zeek, Wireshark

Example Threat Hunting Use Case

Hypothesis:

"Check if any endpoints executed PowerShell connecting to external IP."

Steps:

1. Query SIEM logs
2. Filter PowerShell commands
3. Identify external connections

4. Match with threat intelligence
5. If malicious → investigate and respond